GitLab Basics for CO7201 Individual Project

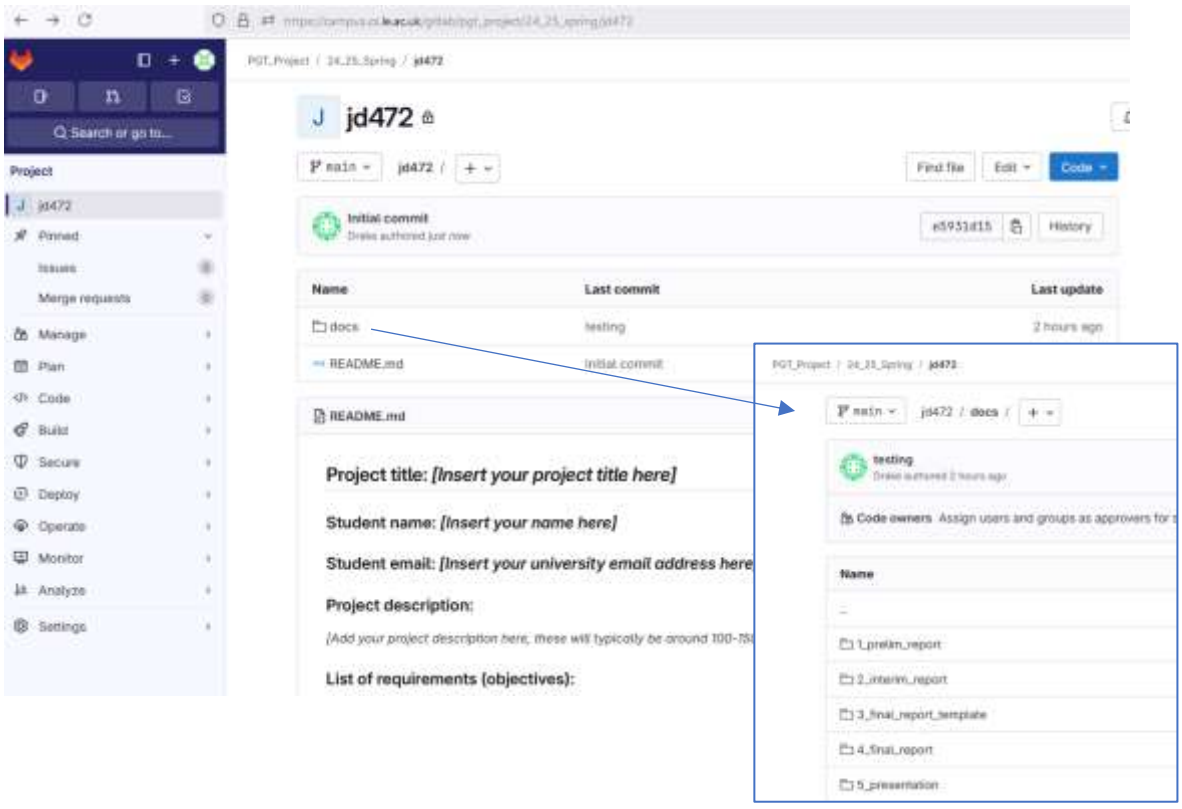
Git is already installed on both the Windows and Linux machines in the lab. If you wish to install Git on your own machine, please follow the installation instructions here: https://carpentries.github.io/workshop-template/#shell.

1. Viewing your GitLab repository online

The address for your personal repository has the following format: https://campus.cs.le.ac.uk/gitlab/pgt_project/24_25_spring/USERNAME_HERE.

Note that you will need to replace the text "USERNAME_HERE" with your own username in this URL. For example, if your username was abc123, the URI for your GitLab repository would be: https://campus.cs.le.ac.uk/gitlab/pgt_project/24_25_spring/abc123.

Please ensure that you only use the GitLab repository that has been set up for you for this module. A default README.md file, and a folder structure for submission of your reports have been created automatically:



2. Getting started with GitLab

In order to start working with this repository, the first step is to clone it to your local system using the following command:

- git clone https://campus.cs.le.ac.uk/gitlab/pgt_project/24_25_spring/USERNAME_HERE.git

Again, please remember to replace the text “USERNAME_HERE” with your own username in this command. If you are struggling with the URI, you can find it under the code dropdown on the web interface for your GitLab repo:



I’ve used GitBash for this example, but this should work under and terminal that you choose. The following series of commands navigate to my Desktop directory, create a new directory called “MyFinalProject”, enter that directory, and then clone a copy of the GitLab repository in that location. Once cloned, the “jd472” directory is now there, containing a “docs” folder and a file called README.md.

```
MINGW64:/z/Desktop Files/MyFinalProject/jd472

jd472@LT-1C1911-001 MINGW64 /
$ cd z:/Desktop\ Files/

jd472@LT-1C1911-001 MINGW64 /z/Desktop Files
$ mkdir MyFinalProject

jd472@LT-1C1911-001 MINGW64 /z/Desktop Files
$ cd MyFinalProject/

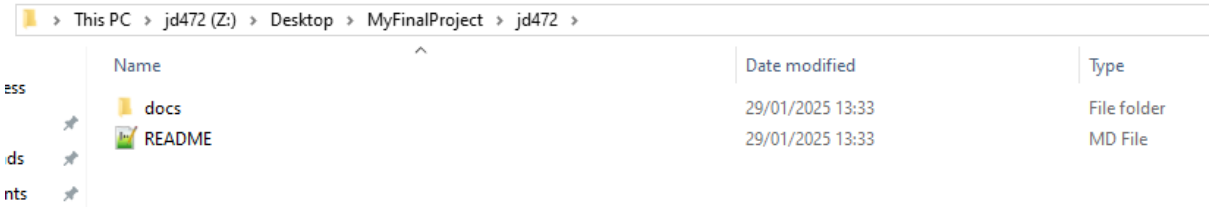
jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject
$ git clone https://campus.cs.le.ac.uk/gitlab/pgt_project/24_25_spring/jd472.git
Cloning into 'jd472'...
remote: Enumerating objects: 84, done.
remote: Counting objects: 100% (75/75), done.
remote: Compressing objects: 100% (71/71), done.
remote: Total 84 (delta 37), reused 0 (delta 0), pack-reused 9 (from 1)
Receiving objects: 100% (84/84), 49.51 KiB | 1.24 MiB/s, done.
Resolving deltas: 100% (38/38), done.

jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject
$ ls
jd472/

jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject
$ cd jd472

jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject/jd472
$ ls
docs/  README.md
```

Although not shown above, you are likely to be asked to enter your username and password as part of the clone operation. We can double check on the windows file system that these have been cloned successfully:



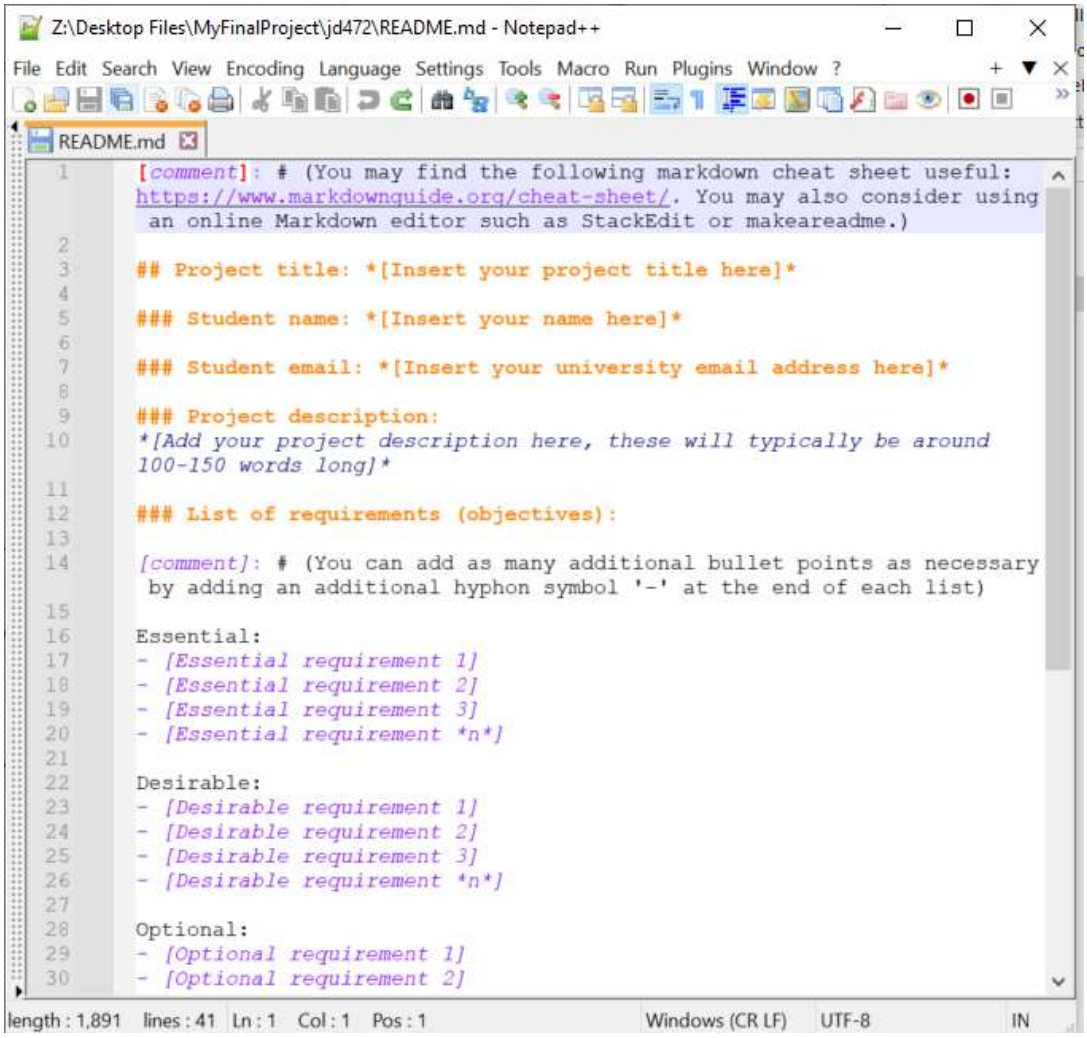
All code development and reports should be saved in this directory directly. As this local folder is a git repository, we can run the git status command at any point to check the current status. At this point, as no changes have been made, so there is nothing to commit:

```
jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject/jd472 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

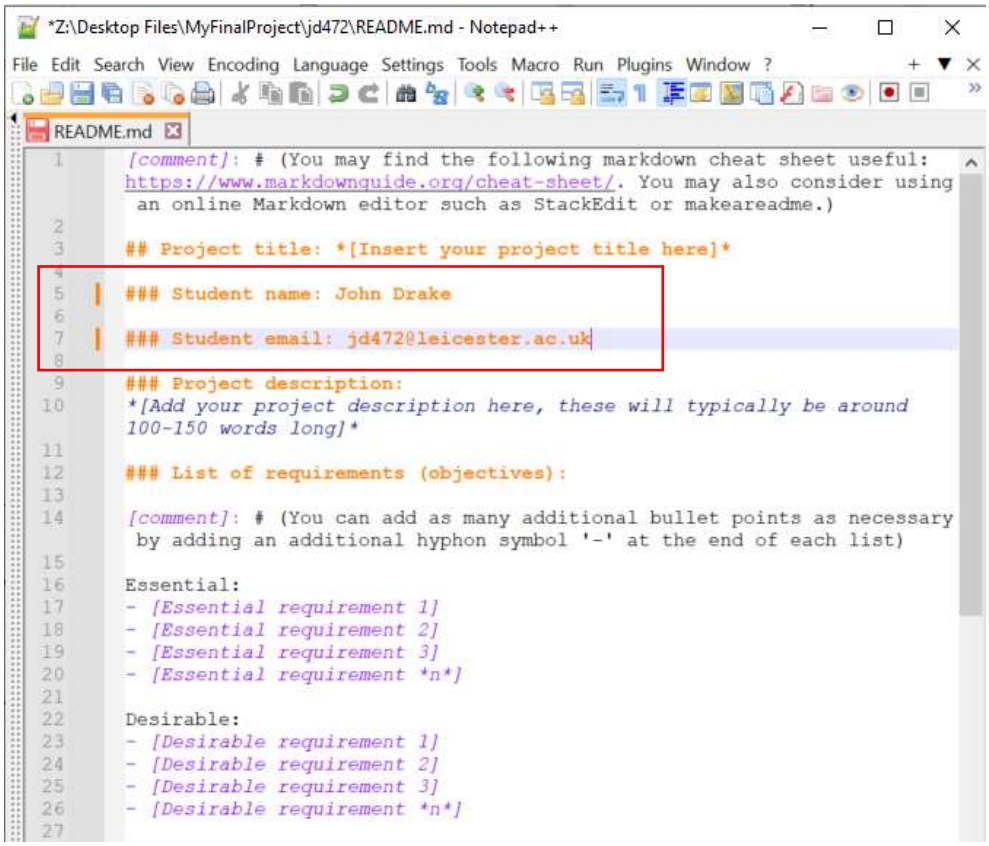
nothing to commit, working tree clean
```

3. Making changes and updating to the server

By default, we have created a README.md (md is a "markdown" file type) template for your Project Description. You may have already noticed that this is displayed on the main page of the GitLab web interface for your repository. Opening the local copy of this file, the contents should look like this:



In order to check that you can update and commit to the GitLab server, we will make some minor changes to this document. As you may not have a project title or full project description written yet, for now we will update the “Student name” and “Student email” information:



Once these changes have been saved, running the “git status” command again should recognise that this file has been modified:

```
jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject/jd472 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

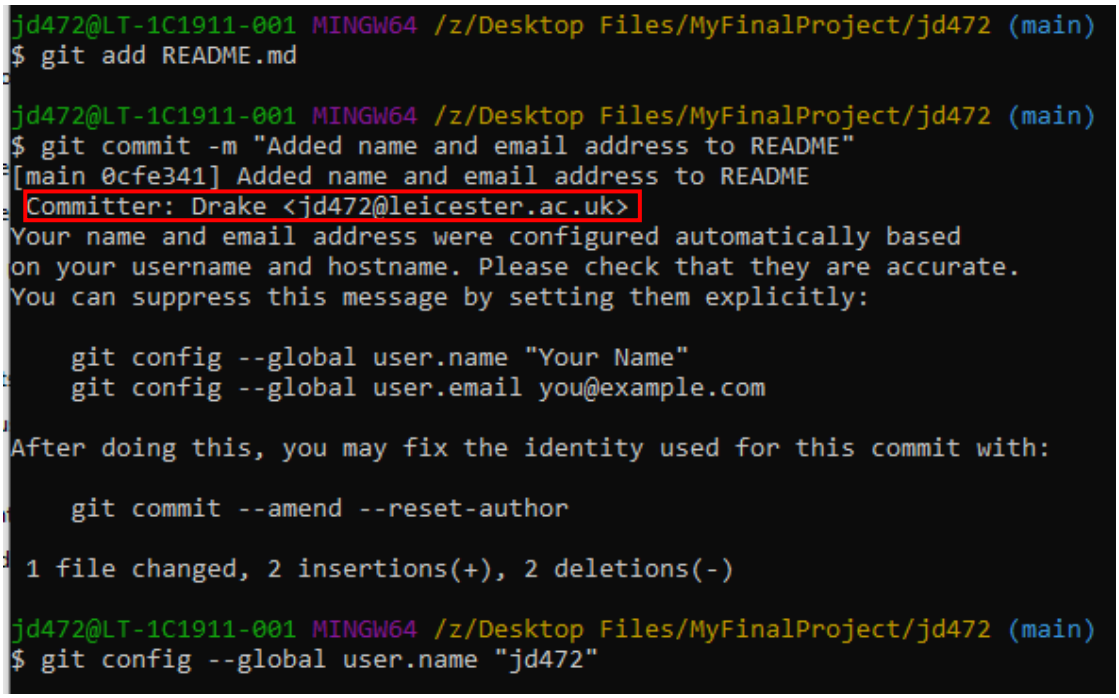
no changes added to commit (use "git add" and/or "git commit -a")
```

Helpfully, git is reminding us that we have not “staged” this file to commit yet. For most purposes in this module, when an update is made to the code, the following three steps need to be completed:

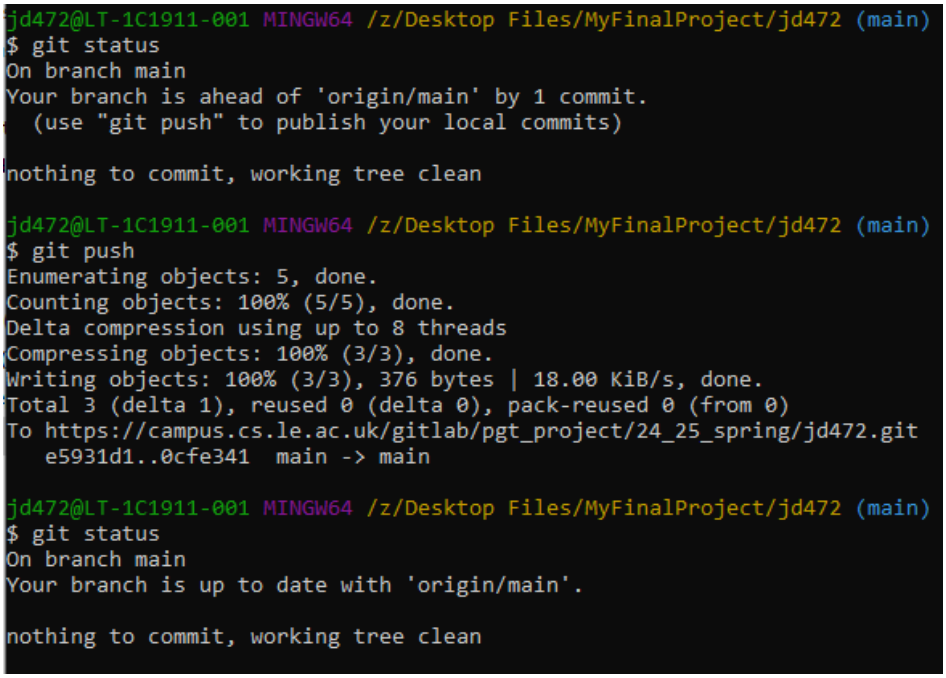
1. The modified (or added) files need to be added to the staging area, so that they are part of the next commit (**git add**).
2. The commit is then made to the local repository (**git commit**). We can think of this as a “save” point for the current state of the repository. This is usually done in combination with the -m option to add a message that describes the changes that have been made in that commit.

3. The updated code is copied from the local repository to the remote repository on the university GitLab server (**git push**).

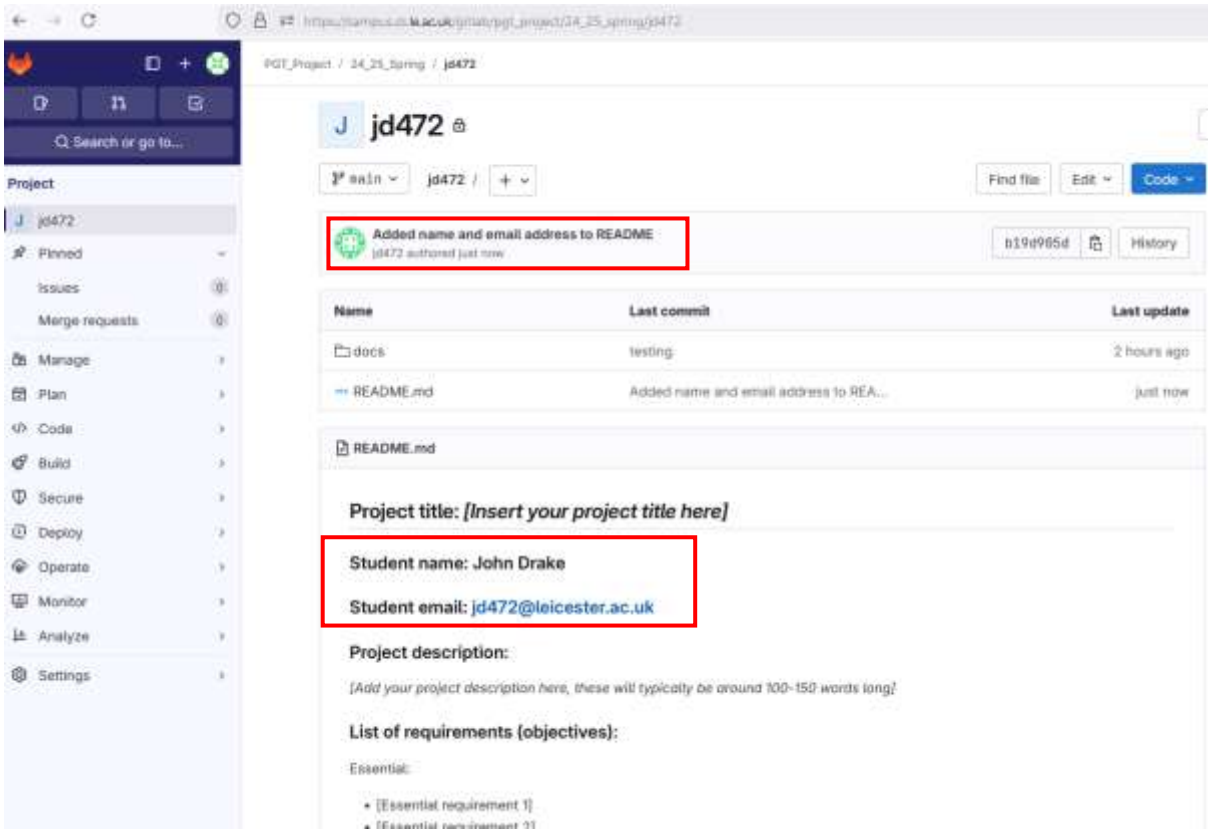
The following two screenshots show this process. First, we add the "README.md" file to the staging area (you can run "git status" after this stage to check), then commit the update with an appropriate message ("Added name and email address to README"). **At this point, double check the details of the Committer and make sure that they are linked to your university account**. If this is not the case, you can update this using the "git config" commands shown.



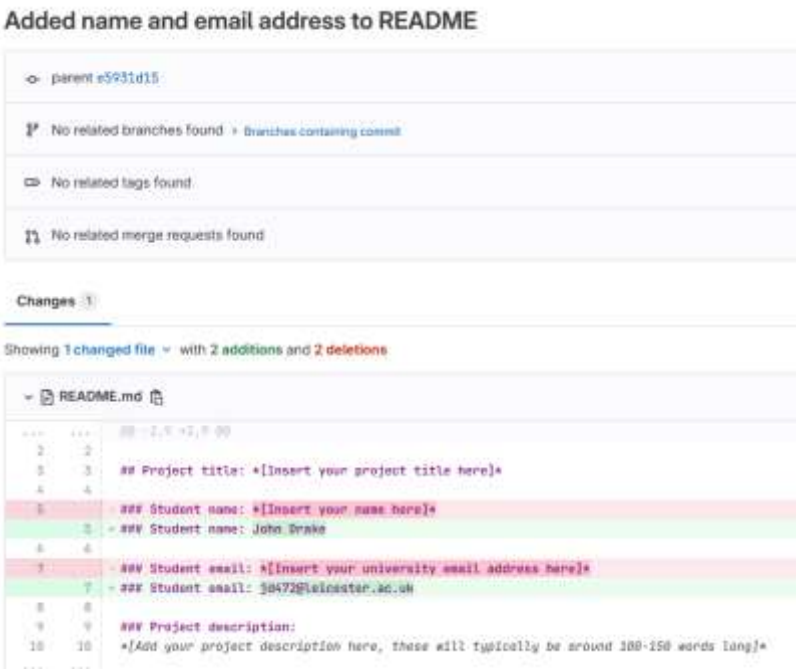
The changes have now been committed to the local repository, but not the server. If we run "git status", we can see that our local repository is one commit ahead (i.e. the server is one commit behind). Running "git push" will push these changes to the remote repository on the server.



Once this is done, we can check this on the GitLab web interface by refreshing the page. The changes made to the README.md are now visible. We can also see the details of the latest commit, and the message added to explain the changes.

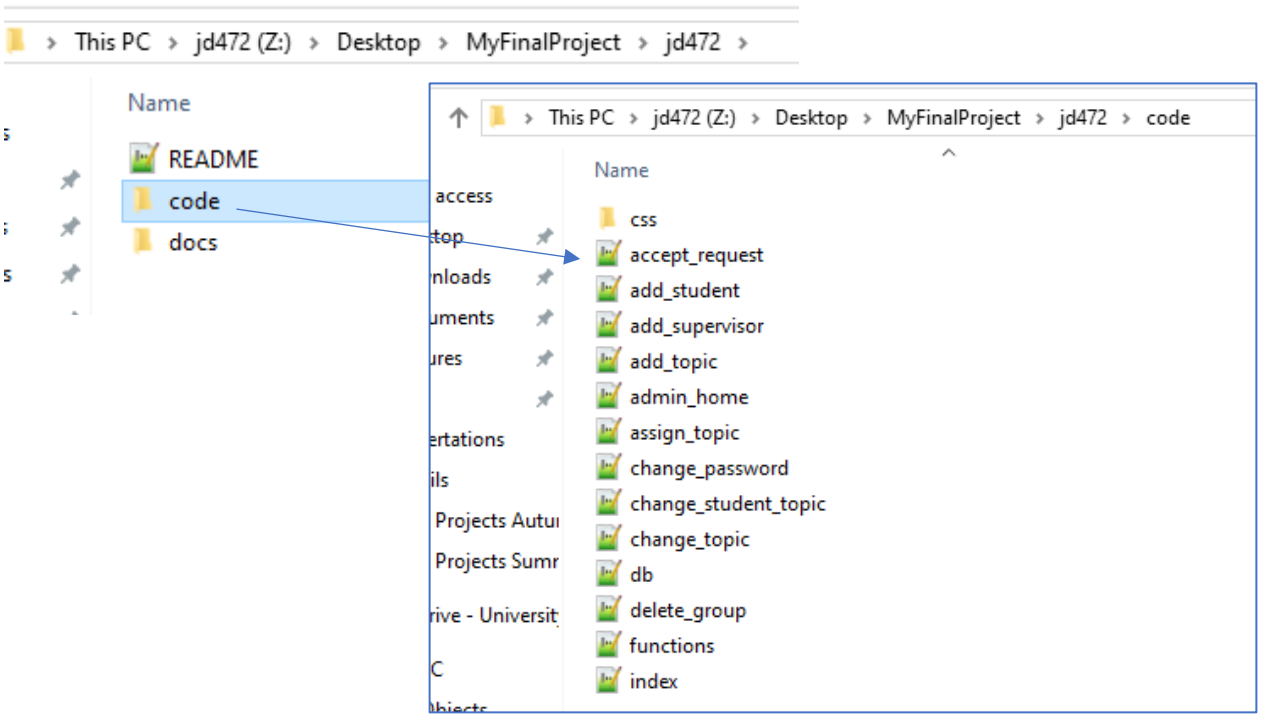


If we click on the name of the commit, it will show a summary of the changes made to each file that were made:

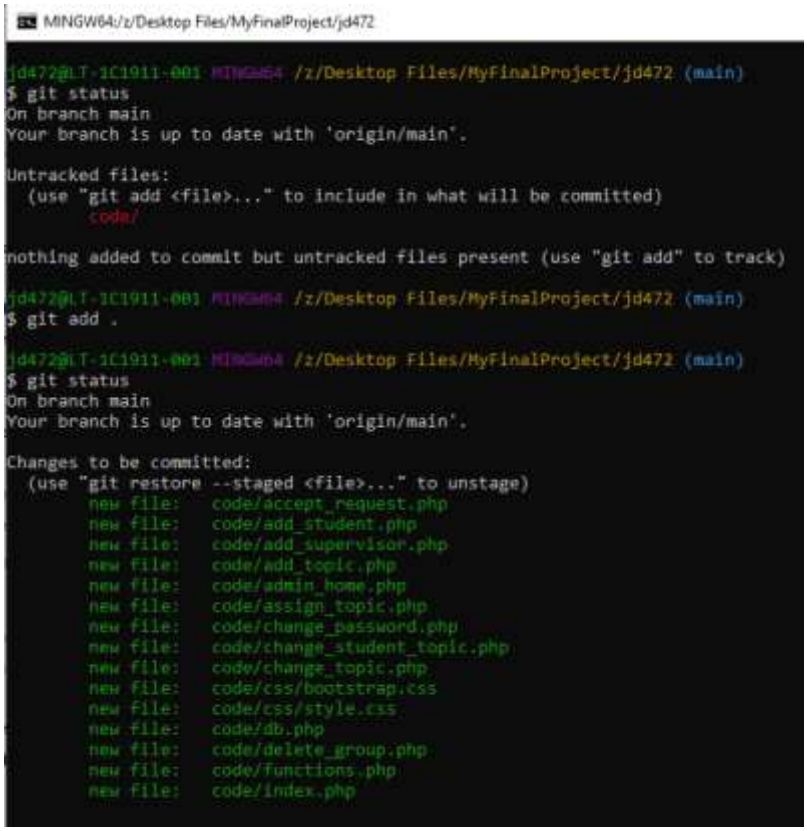


4. Adding multiple files to the repository

In the example above, we added a single file (README.md) to our repository. Once you are into the main development phase of the project, it is likely that you will be updating a large number of files on a regular basis. Below is an example where I have been working on files in a "code" folder within my git repository:



We follow exactly the same process as before (git add, git commit, git push), using a "." (dot, full stop) in the git add command to indicate that all files and folders in the current directory should be added:



```
jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject/jd472 (main)
$ git commit -m "Adding initial codebase with basic functionality"
[main 3091695] Adding initial codebase with basic functionality
 Committer: jd472 <jd472@leicester.ac.uk>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

    git config --global user.name "Your Name"
    git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

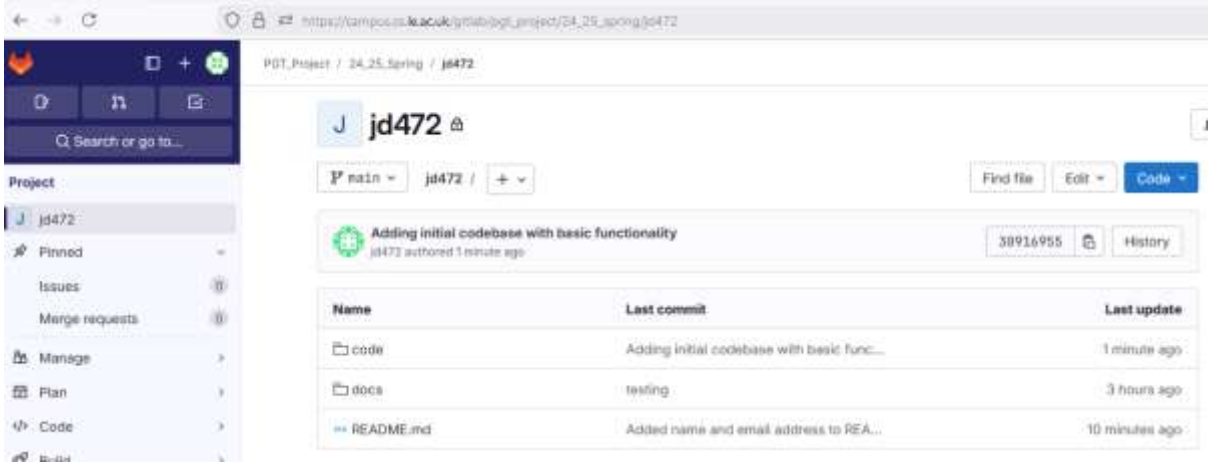
 15 files changed, 8903 insertions(+)
 create mode 100644 code/accept_request.php
 create mode 100644 code/add_student.php
 create mode 100644 code/add_supervisor.php
 create mode 100644 code/add_topic.php
 create mode 100644 code/admin_home.php
 create mode 100644 code/assign_topic.php
 create mode 100644 code/change_password.php
 create mode 100644 code/change_student_topic.php
 create mode 100644 code/change_topic.php
 create mode 100644 code/css/bootstrap.css
 create mode 100644 code/css/style.css
 create mode 100644 code/db.php
 create mode 100644 code/delete_group.php
 create mode 100644 code/functions.php
 create mode 100644 code/index.php

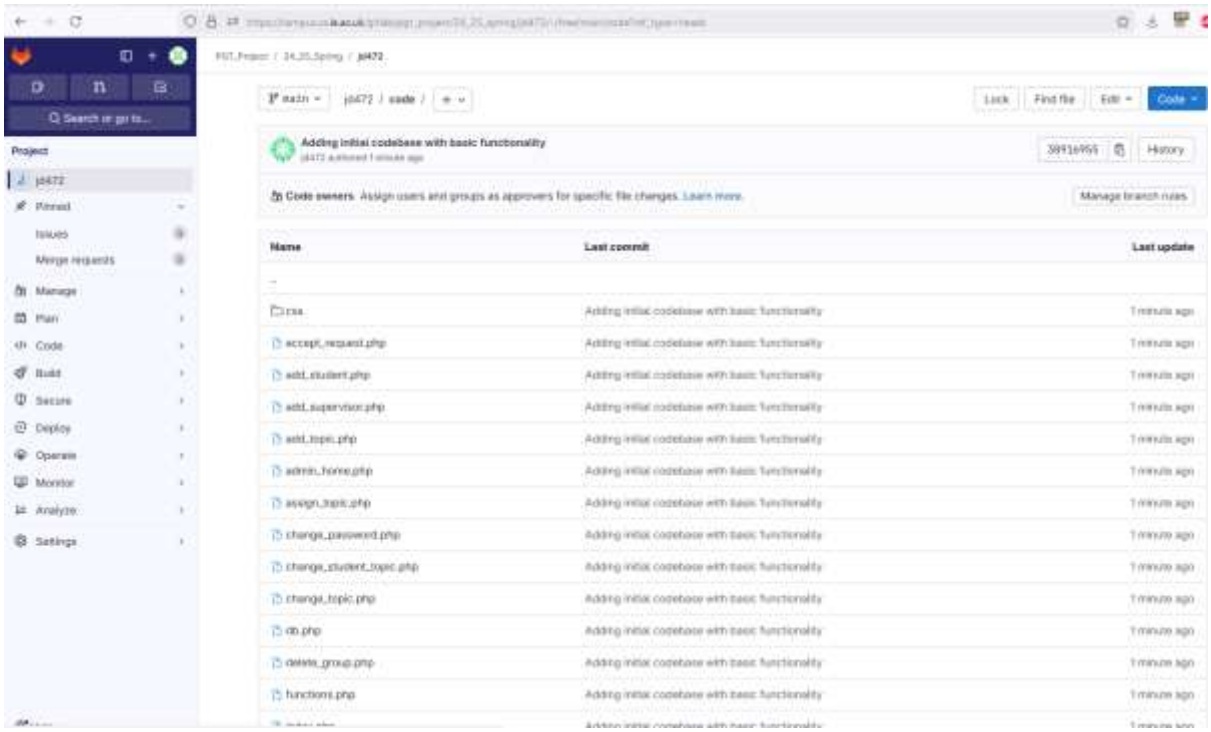
jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject/jd472 (main)
$ git push
Enumerating objects: 20, done.
Counting objects: 100% (20/20), done.
Delta compression using up to 8 threads
Compressing objects: 100% (15/15), done.
Writing objects: 100% (19/19), 29.03 KiB | 990.00 KiB/s, done.
Total 19 (delta 7), reused 13 (delta 4), pack-reused 0 (from 0)
To https://campus.cs.le.ac.uk/gitlab/pgt_project/24_25_spring/jd472.git
   b19d905..3091695  main -> main

jd472@LT-1C1911-001 MINGW64 /z/Desktop Files/MyFinalProject/jd472 (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

As before, we can use the web interface to check that the expected changes have been made to the server:





Further resources for Git:

Online tutorial: https://uniofleicester.github.io/git-novice/index.html

Basics lecture (UG): https://campus.cs.le.ac.uk/teaching/resources/CO7201/documents/Gitlab-basics.pdf